



National Public School
SARJAPUR ROAD

GRADE X (2026-27)

PROJECTS/ HOLIDAY HOMEWORK

SUBJECT	TOPIC	SPECIFICATION
ENGLISH	NELSON MANDELA – Long Walk to Freedom	Write a short speech on freedom inspired by Mandela in 100-120 words. Instructions: Use a proper format of speech writing (greeting, introduction, body, and conclusion). Highlight important values such as courage, forgiveness, equality, and perseverance. Write in neat handwriting on an A4 sheet.
	DUST OF SNOW & FIRE AND ICE	Students are required to prepare a Concept Map on an A3 sheet comparing and contrasting the poems “Dust of Snow” and “Fire and Ice”. The concept map should clearly show: ☐ the central themes of both poems. ☐ the emotions conveyed. ☐ the symbols used by Robert Frost. ☐ the message and tone of each poem. ☐ similarities and differences in Frost’s depiction. Students must creatively illustrate how Robert Frost presents powerful emotions in “Fire and Ice” and how he portrays a subtle, transformative moment in “Dust of Snow.”
2 nd Language KANNADA	ಭಾಗ್ಯಶಿಲ್ಪಿಗಳು ಪತ್ರಲೇಖನ ಪ್ರಬಂಧ	1. ಮೈಸೂರು ಸಂಸ್ಥಾಪಕರಾದ ನಾಲ್ವಡಿ ಕೃಷ್ಣರಾಜ ಒಡೆಯರು ಮತ್ತು ವಿಶ್ವೇಶ್ವರಯ್ಯ ರವರ ಬಗ್ಗೆ ಪರಿಚಯ ಮತ್ತು ಸಾಧನೆಗಳನ್ನು ಚಿತ್ರದೊಂದಿಗೆ ಬರೆಯಿರಿ ಕಾರ್ಯ ಯೋಜನೆಯ ನಿಯಮಗಳು 1. A4 ಹಳೆಯಲಿ ಅಂದವಾಗಿ ಬರೆಯಿರಿ 2. ಮಾಹಿತಿಗೆ ತಕ್ಕಂತೆ ಚಿತ್ರವನ್ನು ಅಂಟಿಸಿ. 3. ಶೀರ್ಷಿಕೆ, ಪರಿವಿಡಿ ಪೀಠಿಕೆ, ವಿಷಯ ವಿವರಣೆ ಉಪಸಂಹಾರ ಗ್ರಂಥ ಋಣ ಈ ಎಲ್ಲಾ ಹಂತಗಳನ್ನು ನಮೂದಿಸಿ. 4. ಪುಟ ಸಂಖ್ಯೆಯನ್ನು ಬರೆಯಿರಿ. 5. ಕಾರ್ಯಯೋಜನೆಯನ್ನು ಸರಿಯಾದ ಸಮಯಕ್ಕೆ ಸಲ್ಲಿಸಿ. 2. ವ್ಯವಹಾರಿಕ ಪತ್ರ 1. ನಿಮ್ಮ ಉರಿನಲ್ಲಿ ಆರೋಗ್ಯ ಕೇಂದ್ರವನ್ನು ಸ್ಥಾಪಿಸುವಂತೆ ಕೋರಿ ಆರೋಗ್ಯ ಸಚಿವರಿಗೆ ಒಂದು ಪತ್ರ ಬರೆಯಿರಿ. ಅಥವಾ

		2. ನಿಮ್ಮ ಊರಿನಲ್ಲಿ ಹಾಜಾಗಿರುವ ರಸ್ತೆಯನ್ನು ದುರಸ್ತಿ ಗಳಿಸಲು ಗ್ರಾಮ ಪಂಚಾಯಿತಿಯವರಿಗೆ ಒಂದು ಮನವಿ ಪತ್ರ ಬರೆಯಿರಿ.
2 nd Language HINDI	रचनात्मक लेखन- विज्ञापन लेखन स्पर्श-भाग- 2. पद्य खंड: 1. कबीर की साखी (साखी) पाठ-01 बड़े भाई साहब (कहानी)	विषय- 1. ऑर्गेनिक मसालों का प्रचार के लिए एक विज्ञापन तैयार कीजिए। 2. अमृत चाय' बनाने वाली कंपनी के उत्पाद के प्रचार-प्रसार हेतु एक आकर्षक और प्रभावशाली विज्ञापन तैयार कीजिए। 3. चुलबुल चॉकलेट' बनाने वाली कंपनी के उत्पाद के प्रचार हेतु एक आकर्षक विज्ञापन तैयार कीजिए। 4. आपको नए साल के अवसर पर अपने परिवार के लिए मुंबई में पाँच दिनों के लिए कमरा बुक करना है। कमरा बुक करने के लिए गोवा के किसी होटल के मैनेजर को एक ई-मेल के लिखिए। 5. महिला सुरक्षा को लेकर पुलिस विभाग को एक ई-मेल के लिखिए। II 1. हिंदी साहित्य के महान कथाकार प्रेमचंद के जीवन-वृत्त, उनके व्यक्तित्व की प्रमुख विशेषताओं तथा उनकी प्रमुख साहित्यिक रचना का क्रमबद्ध वर्णन कीजिए।
2 nd Language FRENCH	PORTFOLIO	1. Présentez-vous - Incluez : nom, âge, ville, classe, langues, loisirs, 2-3 adjectifs de personnalité et vos projets. Utilisez le présent et des phrases simples. 2. Décrivez votre famille- Incluez : nombre de personnes, relations, professions des parents, descriptions avec adjectifs, activités en famille. Faites attention aux accords. 3. La pollution- Incluez : définition, types, causes, effets, solutions et actions personnelles. Utilisez des phrases simples avec des connecteurs comme <i>parce que, donc, mais</i>.

SANSKRIT	रचनात्मक लेखन अनुच्छेद लेखन Portfolio	1. वायु प्रदूषणम् 2. पुस्तकालयः पाठ्य पुस्तक-शेमुशी-2 पाठ1. शुचिपर्यावरणम् (1st& 2nd श्लोक का अर्थ) i संस्कृत भाषा महत्वः ii बाणभट्ट परिचयः
MATHEMATICS	Indian Mathematicians	Title : Indian Mathematicians and Their Contributions ➤ Prepare the project in A4 sheets with a neat presentation. ➤ Include a Cover Page, Acknowledgement, Index, Introduction, Content, Conclusion, and Bibliography. ➤ Write about 4–5 Indian mathematicians and their contributions. ➤ The main content should be 4–5 pages in length. ➤ Use proper headings and subheadings. ➤ Include pictures or illustrations related to mathematicians. ➤ Maintain neat handwriting and proper margins. ➤ Mention all sources used in the Bibliography.
SCIENCE Biology:		Complete Lab Experiments by referring Science lab manual- Experiment 6 (Temporary mount of leaf to show stomata), Experiment 7 (CO ₂ is released during respiration), Experiment 14 (Identification of different parts of embryo in dicot seeds) in your Biology practical record. Ensure that each experiment includes the following components: Aim, Materials Required, Procedure, Observation and Result with Neatly drawn and properly labelled diagrams. All work must be completed systematically and maintained neatly in the Biology record.
Chemistry:		Complete Lab Experiments by referring Science lab manual- Experiment 2: Performing and observing the reactions and classifying them into Combination, Decomposition, Displacement and Double Displacement.
	Sustainable Management of Natural Resources	Instructions: 1. Research and prepare a detailed project on any two natural resources and study the various practices adopted across India for their sustainable management.

		2. Your project must include the following: ○ Importance of the selected natural resources ○ Causes of their depletion ○ Methods and practices used for sustainable management in different parts of India 3. Present your findings in a well-organised and creative manner. You are encouraged to enhance your project by using recyclable and reusable materials wherever possible. 4. Ensure that the content is neat, informative, and supported with relevant examples, pictures, or case studies.
Physics		1. Tracing the path of a ray of light passing through a rectangular glass slab for different angles of prism 2. Determination of focal length for concave and convex mirror 3. Tracing the path of the rays of light through a glass prism
SOCIAL SCIENCE		<u>Economics Project</u> Every student must prepare <u>one project</u> on any one of the following themes: 1. Consumer Awareness 2. Sustainable Development The project should help you understand real-life economic issues, develop analytical thinking, and apply concepts learned in class. Objectives of the Project The project aims to help students: ● Understand economic concepts in real-life situations ● Develop research and analytical skills ● Encourage collaboration and creativity ● Build communication and presentation skills ● Understand the role of economics in solving social problems Students may use primary sources (interviews, surveys, observation) and secondary sources (books, newspapers, internet). Suggested Project Topics (<u>CHOOSE ANY ONE TOPIC</u>) <u>Option 1: Consumer Awareness</u> ● Rights and responsibilities of consumers ● Consumer Protection Act and its importance ● Common consumer problems in your locality ● Role of consumer courts

		● Case study of a consumer complaint ● Awareness about MRP, expiry date, quality marks (ISI, AGMARK, FSSAI) Activity Idea: Conduct a small survey of 10–15 people asking: ● Do they check expiry date? ● Do they know consumer rights? ● Have they ever filed a complaint? Present results using charts or graphs. <u>Option 2: Sustainable Development</u> ● Importance of sustainable development ● Conservation of resources ● Water conservation in your area ● Renewable vs non-renewable resources ● Waste management in your locality ● Role of individuals in protecting the environment <u>Creative Component</u> Students are encouraged to include: ● Posters ● Charts ● Diagrams ● Newspaper clippings ● Photographs ● Case studies Geography: On A4 size sheet, present a write up on “How the resources are independent in nature and the need to develop them in India. Also present it in the form of a Venn Diagram. Democratic politics: Create a powerpoint presentation on “Policies and Politics that strengthens federalism in India.” Maximum slides: 3-4
ARTIFICIAL INTELLIGENCE NCE	Introduction to Machine Learning, AI Bias & Social Impact	Complete the AI course available at https://code.org/oceans. This course is based on the Machine Learning domain of Artificial Intelligence and provides insight into how AI systems are trained to identify and classify different types, shapes, colours, emotions, and other features of ocean creatures.

		After successfully completing the course, enter your name, download your certificate, and upload it to your Digital Portfolio under the subject Artificial Intelligence. Then, answer the following questions briefly: 1. Explain Machine Learning. How is AI trained? 2. Describe AI bias and fairness. What are the social impacts of AI?
	Computer Vision in Artificial Intelligence	To play the game, visit: https://emojiscavengerhunt.withgoogle.com/ This game is based on the Computer Vision domain of Artificial Intelligence. The system displays an emoji, and the participant must quickly find and show a matching real-world object in front of the camera. The AI then attempts to identify the object. The challenge is to correctly find and show eight items within the given time limit in order to win the game. After playing the game, answer the following questions: 1. Were you able to complete the challenge successfully? 2. What strategy did you use to win the game? 3. Was the computer able to correctly identify all the objects you showed? 4. Did the lighting conditions in the room affect the machine's ability to recognize the objects?